

A Hybrid Class-E Topology With Constant Current and Constant Voltage Output for Light EVs Wireless Charging Application

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Abstract—Traditional Class-E circuit features fewer components, high reliability, and soft switching. However, due to the unidirectional excitation, the application of such a topology circuit is limited, i.e., usually used in low-power applications. To relieve this issue, this article proposes a switched-capacitor-based Class-E circuit, to extend the operating power range of the Class-E circuit. With the proposed new circuit, both constant current (CC) and constant voltage (CV) output modes can be achieved by switching the branch once without changing the switching frequency or compensation network. Compared with the existing CC and CV charging strategies, the proposed circuit has the advantages of fewer components, simpler implementation, and higher reliability. Moreover, the studied compensation network can also be extended to other inverter topologies. The working principle of the proposed circuit is described in detail, and the variable zero-voltage switching (ZVS) margin is introduced to make an accurate design of parameters. Subsequently, the influence of high-order harmonic components on the circuit and the sensitivity of parameters are analyzed. Finally, the proposed circuit is simulated and experimented on a 180–450-W prototype with CC and CV characteristics to verify the feasibility of the circuit and the accuracy of theoretical analysis.

Index Terms—Class-E, constant current (CC) and constant voltage (CV) output, higher order harmonics, wireless power transfer (WPT).

I. INTRODUCTION

WIRELESS power transfer (WPT) is a popular power conversion technology that allows power transfer through the magnetic field coupling of the primary and secondary coils. The WPT system provides a lot of exciting features, such as safety, convenience, flexibility, and unnecessary of bulky cable and heavy plugs [1], [2]. These substantial operational metrics improvements make WPT gain more and more attention in both industry and academia. The basic theory of

WPT can be mainly divided into three categories: inductively coupled power transfer (ICPT), magnetically coupled resonant power transfer, and microwave WPT. Nowadays, inductively coupled power transmission has been widely used in wireless charging [3], [4], implantable medical equipment [5], ocean exploration [6], underground mine power supply [7], and other applications [8], [9].

Among different WPT topologies, the full-bridge inverter topology is widely used in medium- and high-power WPT systems [10]. However, there are four active switches needed for the full-bridge inverter, which results in unexpected high cost, complex implementation, and difficult for zero-voltage switching (ZVS) realization [11], [12]. On the other hand, another topology circuit, namely Class-E circuit, is generally used in low power but high-frequency applications. This circuit has only one active switch, which features high efficiency, low cost, and ZVS realization [13]–[15]. Therefore, the Class-E topology circuit is popular in low-power WPT systems. However, due to the characteristic of unidirectional excitation, it is hard for Class-E circuit to achieve a wide power transmission range, i.e., only a few or dozens of watts. In [16], a 6.78-MHz wireless power transmitter is presented based on a reconfigurable Class-E power amplifier for various and multiple device charging. The proposed circuit in [16] has advantages in charging various types of single and multiple devices while maintaining a high efficiency. However, the maximum charging power of this system is merely 39 W, which greatly limits its range of application.

To allow a bidirectional excitation and thus improve the operating power range, a switched-capacitor-based Class-E topology circuit is proposed in this article. The traditional single switch circuits, such as Forward and Flyback, cannot realize bidirectional excitation and thus can only transfer power when the switch is ON or OFF. However, the Class-E circuit in this article can transfer power to the secondary side whether the switch is ON or OFF. In addition, the power transmission level can be increased to several hundred watts and thus greatly improves the power transmission capacity of the circuit. This hybrid topology introduces the S/capacitance and inductance (CL) compensation network and full-bridge rectifier in Class-E circuit as well, which can improve the

Manuscript received November 5, 2020; revised February 2, 2021 and April 9, 2021; accepted May 17, 2021. Date of publication May 24, 2021; date of current version October 1, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 51577118. (Corresponding author: Yong Wang.)

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Digital Object Identifier 10.1109/TTE.2021.3083363

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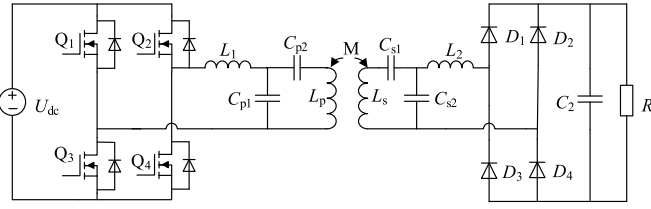


Fig. 1. Double-sided LCC compensation network.

output power and realize the constant current (CC) and constant voltage (CV) output characteristics. For wireless battery charging applications, it is necessary to realize CC and CV output modes for the sake of extending the life span of battery and simplifying the circuit design [17]. Nowadays, CC and CV output modes in WPT systems are generally realized by introducing double-sided LCC compensation network in full-bridge inverter circuits [18]–[20], where CC and CV modes can be switched by changing the operating frequency. In [21], a double-sided LCC compensation network and its tuning method for WPT system is proposed, as shown in Fig. 1, which makes the resonant frequency independent of the operating condition and the coupling coefficient between coils. Nevertheless, this circuit requires more power devices, which leads to larger circuit volume, more resonant circuits, and lower efficiency in medium- and low-power applications. In addition, since the operating frequency of the system needs to be changed while switching between CC and CV modes, and the stability of the system is reduced. Moreover, due to the limitation of industrial, scientific, and medical (ISM) band, the frequency of the system cannot be changed arbitrarily at higher frequency, so this method is not practically available. In [22], two hybrid topologies are proposed using SS and PS compensation or SP and PP compensation for battery charging, as shown in Fig. 2(a) and (b), respectively. Although they can switch CC and CV modes at a fixed frequency, there are three switches in use, which makes the circuit structure more complex. Furthermore, a new resonant circuit is added, which changes the original compensation network. In [25], a full-bridge circuit and a primary-side reconstruction compensation network are adopted, and CC and CV modes can be realized by switching only once, but it uses more passive components and is not suitable for the applications where the reflection impedance of the transmitter is small. In [26], the resonant frequency is set to be about twice the switching frequency to achieve ZVS during the charging process. However, the soft-switching margin is not considered, resulting in a certain deviation between the calculated resonant voltage amplitude and the actual value. In addition, the influence of high-order harmonics in the waveform on the circuit is not considered. Therefore, it is of great significance to find a simple topology, which achieves CC and CV modes switching at a fixed frequency as well.

To allow a bidirectional excitation and thus improve the operating power range, a switched-capacitor-based Class-E topology circuit is proposed in this article. The application of this article is the wireless charging of electric bicycle or electric scooters about 180 W at the 100-kHz switching frequency. Although the experimental frequency is higher than 85 kHz,

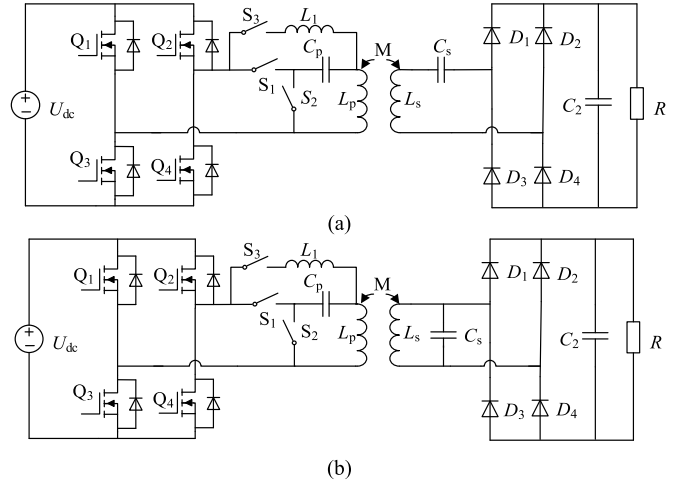


Fig. 2. Two hybrid topologies for battery charging. (a) Hybrid SS and PS compensation. (b) Hybrid SP and PP compensation.

which may cause more radiation effects, it is still within the radiation range specified by the International Commission on Non-Ionizing Radiation Protection (ICNIRP). With the proposed topology circuit, the power can be transferred from the primary side to the secondary side no matter the active switch is ON or OFF. Such a characteristics cannot be realized by traditional single active switch-based topology circuits, i.e., Forward and Flyback. In addition, the compensation network of the switched-capacitor-based Class-E circuit studied in this article is based on the S/CL structure, where the number of power devices is greatly reduced, and only one active switch is used to switch CC and CV modes without changing the compensation network structure, which can also improve the output power range and realize the CC and CV output characteristics. Compared with the traditional full-bridge inverter circuit, the proposed the switched-capacitor-based Class-E circuit has the advantages of high reliability, low cost, simple structure and control, being easy to implement ZVS, and no shoot-through problem [23], [24]. Furthermore, the ZVS margin is taken into consideration, and a more accurate theory is given, where the resonance waveform is decomposed through a Fourier transform, and the influence of high harmonic components on the proposed circuit is considered.

Compared with the traditional Class-E topology, the proposed switched-capacitor-based Class-E inverter can achieve the ZVS operation under a wide range of load variations, e.g., from 7 to 60 Ω in the experiments. Compared with the traditional scheme of CC and CV operations, the switched-capacitor-based Class-E inverter requires less power devices, which helps improve the power density and robustness of the circuits. Due to the compensation network and the design method, the proposed switched-capacitor-based Class-E inverter can achieve higher output power than the traditional Class-E circuit.

The rest of this article is organized as follows. The operation principle of the switched-capacitor-based Class-E circuit and the CC and CV compensation network is described in detail in Section II. The voltage of resonant circuit is modeled and analyzed by the Fourier decomposition, and the parameter

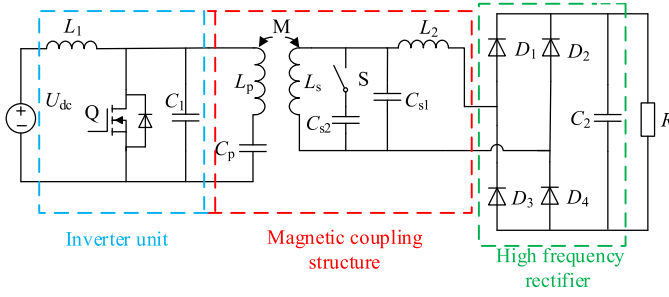


Fig. 3. Switched-capacitor-based Class-E circuit.

TABLE I
COMPARISON OF THE NUMBER OF POWER DEVICES

Number	[21]	[23]	[28]	Proposed
Inductors	2	1	2	2
Capacitors	4	2	5	4
Switch	4	4	4	1
SPDT	0	3	2	1
Total	10	10	13	8

design theory is given in Section III. The switched-capacitor-based Class-E circuit and compensation network are simulated, and the prototype is built for verification in Section IV. Finally, the conclusions are summarized in Section V.

II. ANALYSIS OF A NOVEL CLASS-E CIRCUIT AND COMPENSATION NETWORK

A. Principle of Main Circuit

One traditional method to realize CC and CV output is through a double-sided LCC compensation network [20], as shown in Fig. 1. Another two methods are through hybrid topologies using either SS and PS compensation or SP and PP compensation [22], as shown in Fig. 2. The former requires variable frequency control, which reduces the stability of the system, whereas the latter needs more power devices and changes the compensation network as well.

In order to solve the problems mentioned above, a switched-capacitor-based Class-E circuit is proposed in this article, as shown in Fig. 3. The primary side of the proposed topology is a Class-E circuit (also called inverter unit), and the secondary side is a CL compensation network (also called magnetic coupling structure) followed by a full-bridge rectifier circuit (also called high-frequency rectifier). The proposed topology can achieve CC and CV output by switching the branch once without changing the switching frequency or compensation network. Moreover, it can realize soft switching and meets the needs of medium- and low-power applications. Table I shows the comparison of the number of power devices used in the circuits between the solutions from the references and the proposed switched-capacitor-based Class-E circuit. It is shown in Table I that this proposed switched-capacitor-based Class-E circuit has fewer power devices.

The inverter unit consists of L_1 , Q , and C_1 , which converts the input dc voltage into high-frequency ac voltage. The magnetic coupling structure consists of L_p , C_p , L_s , C_{s1} , C_{s2} ,

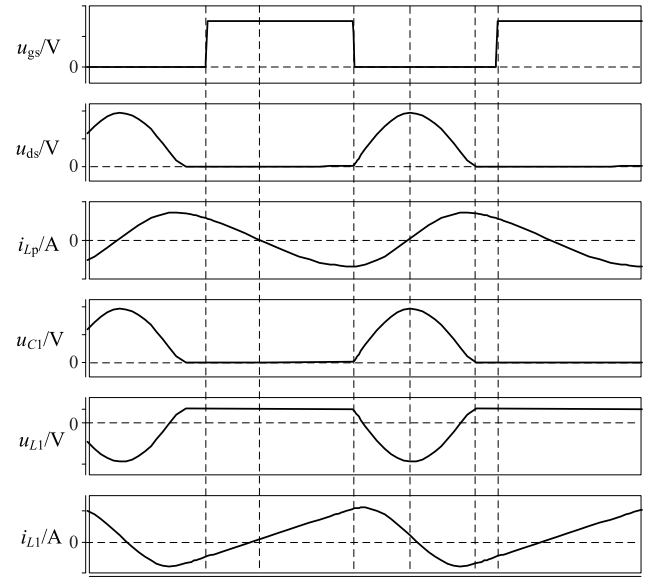


Fig. 4. Key waveforms of inverter unit.

S , and L_2 , which transfers energy from the transmitter to the receiver through the magnetic field. L_p and L_s are transmitting coil and receiving coil, respectively, and the switch S is used to switch CC and CV mode. When the switch S is OFF, the secondary compensation capacitor is only composed of C_{s1} , and CC mode can be realized. When the switch S is ON, C_{s1} is paralleled to C_{s2} , and operation mode changes to CV. The high-frequency rectifier consists of four diodes, which converts high-frequency ac voltage into dc voltage to supply the load.

The key waveforms of inverter unit are shown in Fig. 4, where u_{gs} is the gate-source voltage of Q , u_{ds} is the drain-source voltage of Q , i_{Lp} is the current through the transmitting coil L_p , u_{C1} is the voltage waveform on capacitor C_1 , which is also the input voltage of resonance circuit, and u_{L1} and i_{L1} are voltage and current waveforms of inductor L_1 , respectively.

As can be seen from Fig. 4, u_{gs} rises after the voltage u_{ds} drops to zero, so the active switch realizes the ZVS turn-on.

B. Analysis of CC Output Mode

The magnetic coupling structure of Class-E circuit is analyzed using the T-type equivalent network, as shown in Fig. 5.

As shown in Fig. 5, the loosely coupled transformer is replaced by a mutual inductance model. The primary inductance L_p is divided into L_{p1} and L_{p2} , and the secondary compensation capacitor is replaced by C_s whose value is determined by the state of the switch S . When the switch S is turned off, C_s is equal to C_{s1} . When the switch S is turned on, C_s is equal to the equivalent capacitance of C_{s1} and C_{s2} in parallel. Input voltage u_{in} is expressed as sine wave (see Section III for detailed analysis), the load is set as a purely resistive load R_{eq} , and the output voltage and current of the compensation network are U_o and I_o , respectively. Since the capacitor C_1 is directly connected to the voltage source u_{in} in parallel, the external characteristics of the compensation network are not affected, and the capacitor C_1 is used to adjust the soft switching of the circuit.

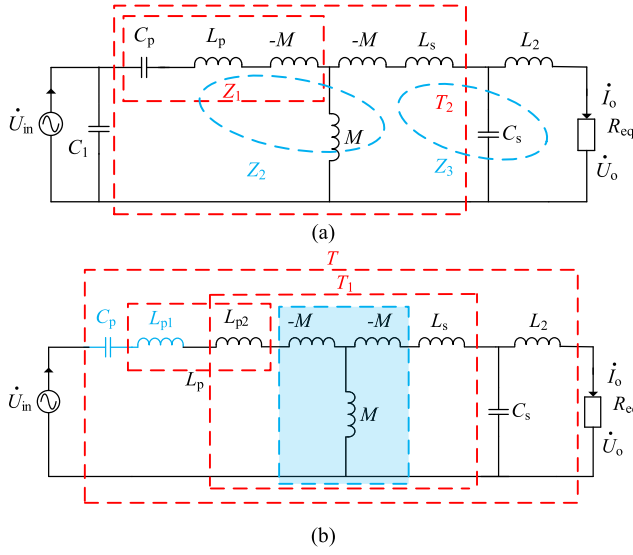


Fig. 5. T-type equivalent network. (a) Equivalent network. (b) Separated inductance network.

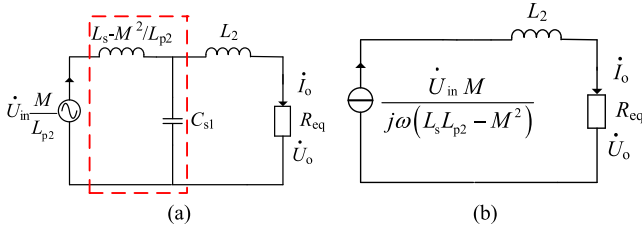


Fig. 6. Equivalent model in CC mode. (a) Intermediate equivalent model. (b) Simplified equivalent model.

Assume that the circuit must work at switching frequency f to realize CC output. If the primary compensation capacitor C_p and inductance L_{p1} also resonate at frequency f , the equivalent impedance of the two components is zero. Thus, based on Thévenin's and Norton's theorems, the T_1 network in Fig. 5 can be converted to an inductance, as shown in Fig. 6(a). In addition, if the equivalent inductance $L_s - M^2/L_{p2}$ and C_{s1} also resonate at frequency f , the structure shown in Fig. 6(b) can be obtained.

As can be seen from Fig. 6(b), the proposed topology is eventually converted to a CC source flowing through inductor L_2 and load R_{eq} , and thus, the CC output mode is realized. From the above analysis, it can be concluded that the condition for the circuit to achieve CC output is

$$\begin{cases} \omega L_{p1} = \frac{1}{\omega C_p} \\ \omega \left(L_s - \frac{M^2}{L_{p2}} \right) = \frac{1}{\omega C_{s1}} \end{cases} \quad (1)$$

where $\omega = 2\pi f$. Thus, when the operating frequency is f , the load-independent output current can be calculated as follows:

$$i_0 = \frac{\dot{U}_{in} M}{j\omega (L_s L_{p2} - M^2)}. \quad (2)$$

It can be concluded from (2) that the output current of the CC output mode is related to the power U_{in} , mutual inductance M and inductance L_s and L_{p2} . Among them, L_{p2} can be divided from the inductance L_p artificially, so this modeling method greatly increases the design freedom of the circuit.

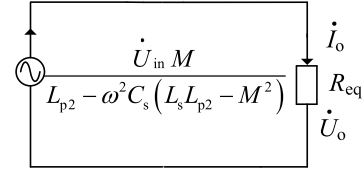


Fig. 7. Equivalent model in CV mode.

The load-independent output current can also be expressed by coupling coefficient k and quality factor Q_{Lp} and Q_{Ls} as follows:

$$i_0 = \frac{\dot{U}_{in} k}{(\alpha - k^2) r \sqrt{Q_{Lp} Q_{Ls}}}. \quad (3)$$

It can be seen from (3) that the output current increases with the growth of coupling coefficient k and decreases with the growth of quality factor Q_{Lp} and Q_{Ls} .

C. Analysis of CV Output Mode

In the CV mode, the switch S is turned on, and thus, C_s is equal to the equivalent capacitance of C_{s1} and C_{s2} in parallel. The circuit is working at switching frequency f , so the equivalent network of the compensation network is still the one shown in Fig. 6(a). From Fig. 6(a), if the parallel combination of inductance $L_s - M^2/L_{p2}$ and capacitor C_s resonates with L_2 at frequency f , the equivalent circuit shown in Fig. 7 can be obtained.

As can be seen from Fig. 7, the final equivalent circuit can be represented as a voltage source in series with load R_{eq} , so the voltage on the load is independent of the resistance, which proves that the CV output mode can be realized. With the above analysis, it can be concluded that the condition for the circuit to achieve CV output is

$$\begin{cases} C_s = C_{s1} + C_{s2} \\ j\omega \left(L_s - \frac{M^2}{L_{p2}} \right) // \frac{1}{j\omega C_s} + j\omega L_2 = 0 \end{cases} \quad (4)$$

where $\omega = 2\pi f$. Thus, when the operating frequency is f , the load-independent output voltage can be calculated as

$$\dot{U}_o = \frac{\dot{U}_{in} M}{L_{p2} - \omega^2 C_s (L_s L_{p2} - M^2)}. \quad (5)$$

In addition, for the T-type equivalent circuit shown in Fig. 6(a), in order to realize zero phase angle (ZPA) in CC mode, the following relationship should be satisfied:

$$j\omega \left(L_s - \frac{M^2}{L_{p2}} \right) = j\omega L_2 = -\frac{1}{j\omega C_{s1}}. \quad (6)$$

Derived from (4) and (6), the expression for the compensation capacitance C_s can be rewritten as follows:

$$C_s = \frac{L_s + L_1 - \frac{M^2}{L_{p2}}}{\omega^2 L_1 \left(L_s - \frac{M^2}{L_{p2}} \right)} = \frac{2}{\omega^2 L_1}. \quad (7)$$

From (1) and (7), it can be concluded that the compensation capacitance C_s of CV mode is twice as that of CC mode C_{s1} , i.e., C_{s1} is equal to C_{s2} .

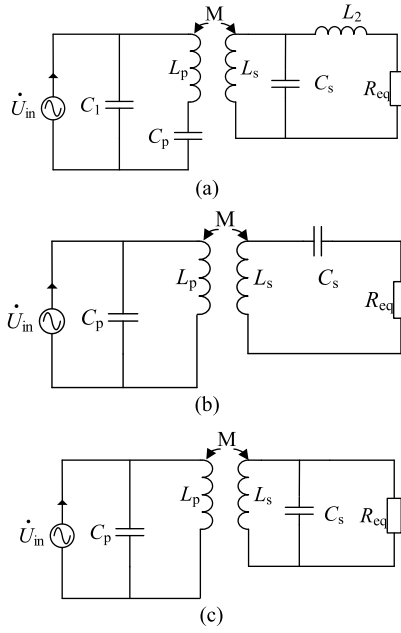


Fig. 8. Three kinds of compensation network diagram. (a) Proposed. (b) PS compensation network. (c) PP compensation network.

D. Power Comparison of the Compensation Network

The proposed Class-E circuit is compared with the traditional PS and PP compensation, as shown in Fig. 8.

The proposed compensation network adopts the parameter design method described in this article, while the PS and PP compensation network adopts the traditional parameter design method. The theoretical output power of three compensation networks can be expressed as follows:

$$P_a = \left[\frac{2U_{in}M}{\pi \omega (L_s L_p - M^2)} \right]^2 R_{eq} \quad (8)$$

$$P_b = \frac{\omega^2 M^2 R_{eq} U_{in}^2}{(R_{eq} R_p + \omega^2 M^2)^2 + \omega^2 L_p^2 R_{eq}^2} \quad (9)$$

$$P_c = \frac{M^2 L_s^2 R_{eq} U_{in}^2}{(M^2 R_{eq} + L_s^2 R_s)^2 + \omega^2 L_s^2 (L_p L_s - M^2)^2} \quad (10)$$

Under the same parameters, the power curves of three compensation networks are plotted, as shown in Fig. 9.

It can be seen from Fig. 9 that the output power of the proposed circuit is higher than that of the traditional PS and PP compensation.

III. PARAMETER ANALYSIS

A. High-Order Harmonics Analysis

Due to the unique structure of Class-E circuit, the waveform of an equivalent input voltage source is different from that of full-bridge circuit. When the switch Q or its body diode is turned on, u_{C1} is clamped on zero. When the switch Q works at the blocking state, u_{C1} is determined by the resonance network.

It can be seen from Fig. 4 that the sum of u_{C1} and u_{L1} is equal to U_{dc} when the switch Q is turned off. Therefore, the voltage waveform on L_1 is analyzed first, as shown in Fig. 10.

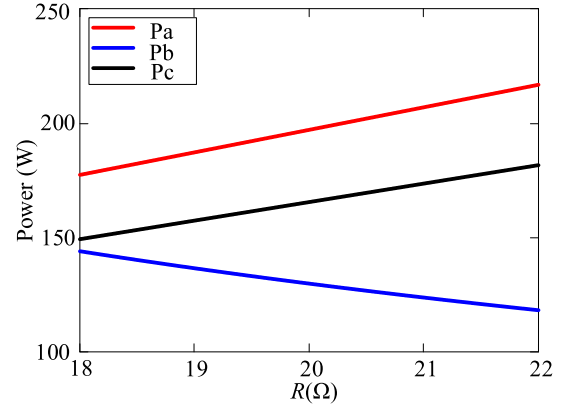


Fig. 9. Power curves of three compensation networks.

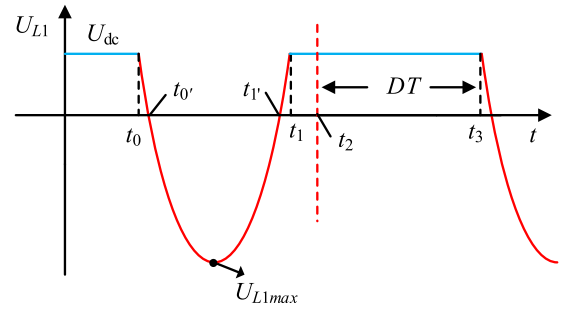


Fig. 10. Voltage waveform on L_1 .

To simplify the analysis, a sinusoidal half-wave can be used to replace the voltage waveform of L_1 , when the switch Q is turned off. Setting t_0 as the starting point, the expression of voltage waveform can be written as follows:

$$\begin{cases} u_{L1}(t) = U_{dc} - A \sin \omega_1 t \\ \omega_1 = 2\pi \frac{1}{t_1 - t_0} \end{cases} \quad (11)$$

where ω_1 and A are the angular frequency and the amplitude of the sinusoidal half-wave, respectively. From Fig. 10, the duration of each segment and the angular frequency ω_1 can be calculated as follows:

$$\begin{cases} t_1 - t_0 = (1 - D - ZV S_m)T \\ t_2 - t_1 = ZV S_m \\ t_3 - t_2 = DT \end{cases} \quad (12)$$

$$\omega_1 = \pi \times \frac{1}{(1 - D - ZV S_m)T} \quad (13)$$

where D is the duty cycle, $ZV S_m$ is the ZVS margin, and T is the operating cycle. Different from the conclusion in [28] that the angular frequency ω_1 of u_{in} is 1.3 times or 2 times of the angular frequency ω corresponding to the switching frequency f , which is not accurate, ω_1 changes as $ZV S_m$ or D . The introduction of $ZV S_m$ makes the model more accurate.

In steady state, the average value of the voltage across inductor L_1 for an operating cycle is zero, which is also called volt-second balance. From Fig. 10, the volt-second balance equation for L_1 can be calculated as follows:

$$|U_{dc}(D + ZV S_m)T| = \left| \int_{t_0}^{t_1} (U_{dc} - A \sin \omega_1 t) dt \right| \quad (14)$$

Substituting (12) and (13) into (14), the value of A can be calculated as

$$A = \frac{U_{dc} T \omega_1}{2} = \frac{U_{dc} \pi}{2(1 - D - ZV S_m)} \quad (15)$$

where A is also the maximum blocking voltage value of the switch Q . According to (11) and (15), the power supply voltage $u_{in}(t)$ of the resonant circuit can be obtained as

$$u_{in}(t) = u_{cl}(t) = \begin{cases} A \sin \omega_1 t & (t_0 \leq t < t_1) \\ 0 & (t_1 \leq t \leq t_3) \end{cases} \quad (16)$$

In order to accurately calculate the output voltage and current, $u_{in}(t)$ is the Fourier decomposition to obtain the fundamental amplitude, and the expression of the Fourier decomposition of (16) can be computed as

$$u_{in}(t) = a_0 + a_1 \cos \omega t + b_1 \sin \omega t + \sum_{k=2}^{\infty} (a_k \cos k\omega t + b_k \sin k\omega t) \quad (17)$$

where a_0 , a_1 , and b_1 can be written as

$$\begin{cases} a_0 = \frac{1}{T} \int_0^{t_1} A \sin \omega_1 t dt = \frac{2A}{T\omega_1} \\ a_1 = \frac{2}{T} \int_0^{t_1} A \sin \omega_1 t \cos \omega t dt \\ = \frac{2A\omega_1}{(\omega_1^2 - \omega^2)T} (\cos \omega t_1 + 1) \\ b_1 = \frac{2}{T} \int_0^{t_1} A \sin \omega_1 t \sin \omega t dt \\ = \frac{2A\omega_1}{(\omega_1^2 - \omega^2)T} \sin \omega t_1 \end{cases} \quad (18)$$

After the Fourier decomposition, $u_{in}(t)$ can also be written as

$$u_{in}(t) = A_0 + A_1 \cos(\omega t + \varphi) + \sum_{k=2}^{\infty} A_k \cos(k\omega t + \varphi_k) \quad (19)$$

where $A_0 = a_0$, $A_1 = (a_1^2 + b_1^2)^{1/2}$, and $\varphi = \arctan(-b_1/a_1)$.

The Fourier decomposition diagram of resonant voltage $u_{in}(t)$ in the case of 100-V output voltage and 100-kHz switching frequency is obtained through simulation, as shown in Fig. 11. The voltage u_{in} is the input voltage of the resonant circuit, which is also the voltage on capacitor C_1 .

It can be seen from Fig. 11 that the amplitudes of the fundamental wave, second harmonic, and third harmonic are 169.8, 99.95, and 31.28 V, respectively, and the high-order harmonics are all less than 10 V. The second and third harmonics should be considered for accuracy due to their large amplitude. In the design of S/CL compensation network, it is hoped that the compensation network can significantly suppress the higher harmonics.

According to Fig. 5, the impedance of the compensation network Z_1 and Z_2 is calculated as follows:

$$\begin{cases} Z_1 = j\omega(L_p - M) + \frac{1}{j\omega C_p} \\ Z_2 = \frac{Z_1 \times j\omega M}{Z_1 + j\omega M} \end{cases} \quad (20)$$

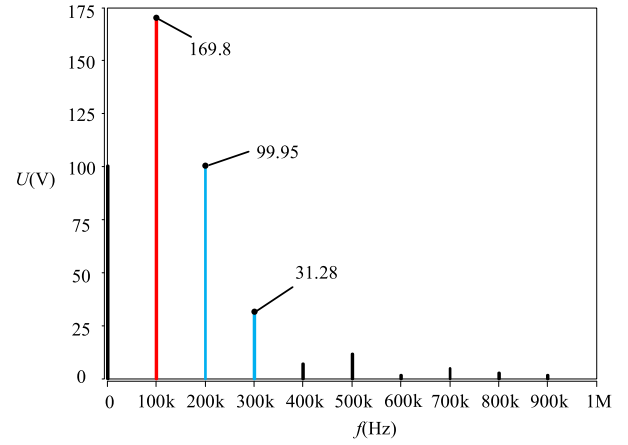


Fig. 11. Fourier decomposition diagram of resonant voltage.

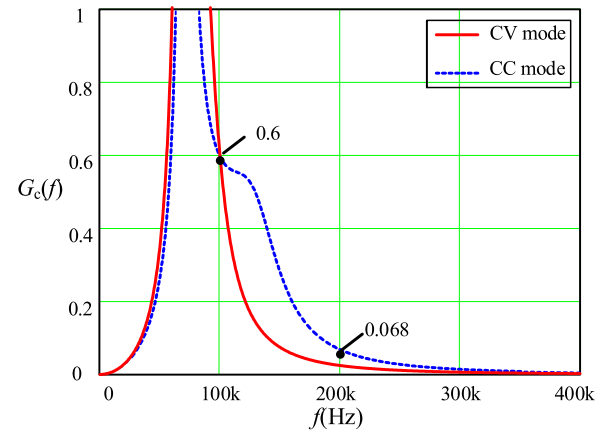


Fig. 12. Curve of gain G_c versus frequency f .

Then, the equivalent impedance of the T_1 network after being connected in series with C_p and L_{p2} Z_{T2} can be expressed as follows:

$$\begin{aligned} Z_{T2} &= j\omega L_{p2} + \frac{1}{j\omega C_p} + Z_{T1} \\ &= \left[j\omega(L_p - M) + \frac{1}{j\omega C_p} \right] // j\omega M + j\omega(L_s - M) \\ &= j\omega \frac{L_s + \omega^2 C_p (M^2 - L_s L_p)}{1 - \omega^2 L_p C_p} = j\omega L_{T2} \end{aligned} \quad (21)$$

where Z_{T1} is the equivalent impedance of the T_1 network. Next, the equivalent impedance of the whole compensation network Z_T can be expressed as follows (22), as shown at the bottom of the next page, where Z_3 is the equivalent impedance of Z_{T2} in parallel with C_s . The voltage gain G_c of the compensation network can be written as

$$G_c = \left| \frac{\dot{U}_o}{\dot{U}_{in}} \right| = \left| \frac{R_{eq}}{Z_T + R_{eq}} \times \frac{Z_2 \times Z_3}{Z_1 \times Z_{T2}} \right|. \quad (23)$$

With (23), the curve of gain G_c versus frequency f is obtained by the Mathcad, as shown in Fig. 12.

It can be seen from Fig. 12 that for fundamental wave, the gain in CC and CV modes is both 0.6; for second harmonic, the gain in CC and CV modes is 0.068 and 0.024, respectively;

and for third harmonic, the gain in CC and CV modes is 0.014 and 0.0062, respectively. Therefore, the designed compensation network can significantly suppress the high-order harmonics. In addition, the gain of the second and third harmonics is low enough to be ignored when calculating the output voltage.

Assuming the ZVS is achieved and considering the conduction losses on the transmitting and receiving coils, switching devices, and the diodes, the efficiency of the proposed WPT system can be generally expressed as

$$\eta = \frac{P_o}{P_o + I_p^2 r_p + I_s^2 r_s + I_{rms}^2 r_{ds} + 2U_d I_d} \quad (24)$$

where I_p , I_s , I_{rms} , and I_d are the currents of transmitting coil, receiving coil, switch Q, and diode, respectively, r_p , r_s , and r_{ds} are the resistance of transmitting coil, receiving coil, and switch Q, respectively, U_d is the forward voltage drop of the diode, and P_o is the output power, which can be obtained from (8).

According to the equivalent circuit of Fig. 8(a), the currents I_p , I_s , I_{rms} , and I_d can be obtained as

$$\begin{cases} I_p = \frac{\dot{U}_{in}}{Z_p + \frac{\omega^2 M^2}{Z_s}} \\ I_s = \frac{j\omega M Y_p \dot{U}_{in}}{Z_s + \frac{\omega^2 M^2}{Z_p}} \\ I_{rms} = \sqrt{\frac{1}{T} \int_0^{DT} i_s(t) dt} \\ I_d = \frac{I_s}{j\omega C_s(j\omega L_2 + R_{eq}) + 1} \end{cases} \quad (25)$$

where $i_s(t)$ is the current on the switch Q and Y_p is the reciprocal of Z_p , Z_p , and Z_s represent the impedance of the transmitting circuit and the receiving circuit, which can be expressed as follows:

$$\begin{cases} Z_p = r_p + j\omega L_p + \frac{1}{j\omega C_p} \\ Z_s = r_s + j\omega L_s + \frac{R_{eq} + j\omega L_2}{j\omega C_s(R_{eq} + j\omega L_2) + 1} \end{cases} \quad (26)$$

B. Analysis of the Relationship Between the Parameters

A full-bridge diode rectifier with C filter is used for rectification, the load current I_b and voltage U_b can be calculated as follows:

$$\begin{cases} I_b = \frac{2}{\pi} I_o \\ U_b = \frac{\pi}{4} U_o \end{cases} \quad (27)$$

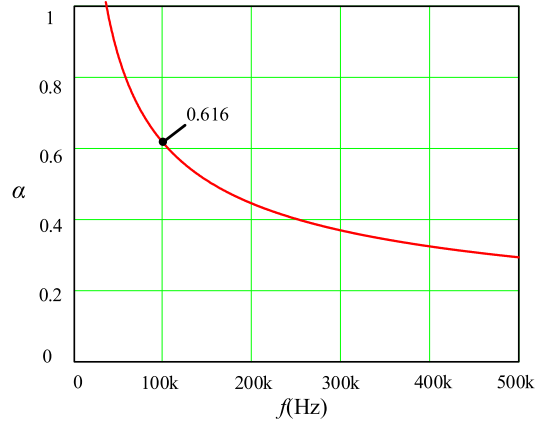


Fig. 13. Curve of α with frequency.

Substituting (2), (5), and (19) into (27), I_b and U_b can be obtained as

$$\begin{cases} I_b = \frac{2A_1 M}{\pi \omega (L_s L_{p2} - M^2)} \\ U_b = \frac{\pi A_1 M}{4[\omega^2 C_s (L_s L_{p2} - M^2) - L_{p2}]} \end{cases} \quad (28)$$

By letting $L_{p2} = \alpha L_p$, and $k = M/(L_p L_s)^{1/2}$, the expression of ω with α , k , and L_p can be obtained as follows:

$$\omega = \frac{A_1^2 k^2}{2U_b I_b L_p \alpha (\alpha - k^2)}. \quad (29)$$

Therefore, when α is given, the condition of $k^2 < \alpha < 1$ needs to be met, and α can be expressed as

$$\alpha = \frac{k^2 + \sqrt{k^4 + \frac{A_1^2 k^2}{\pi U_b I_b L_p f}}}{2}. \quad (30)$$

To charge a 48-V, 20-Ah battery, a 100-V-input Class-E circuit with 3-A output current in CC mode and 60-V output voltage in CV mode is built, and its compensation network parameters can be calculated through (1), (4), (7), and (28). Choosing a pair of transmitting and receiving coils arbitrarily, such as $L_p = 20 \mu\text{H}$, $L_s = 30 \mu\text{H}$, and coupling coefficient $k = 0.25$, the scale factor α versus f is drawn in Fig. 13 by using (30).

It can be seen from Fig. 13 that when the switching frequency f is 100 kHz, the scale factor α is 0.616, and α decreases with the increase of switching frequency.

Fig. 14(a) shows the relationship between the frequency f and the coupling coefficient k with different scale factors α , and Fig. 14(b) shows the relationship between the frequency f and the transmitting coil L_p with different coupling coefficients k .

As shown in Fig. 14, with a certain coupling coefficient, if the operating frequency grows, the scale factor α will increase,

$$Z_T = j\omega L_{T2} // \frac{1}{j\omega C_s} + j\omega L_2 = j\omega \frac{L_2(1 - \omega^2 L_p C_p) + (1 - \omega^2 L_2 C_s)[L_s + \omega^2 C_p(M^2 - L_p L_s)]}{1 - \omega^2(L_p C_p + L_s C_s) - \omega^4 C_s C_p(M^2 - L_p L_s)} \quad (22)$$

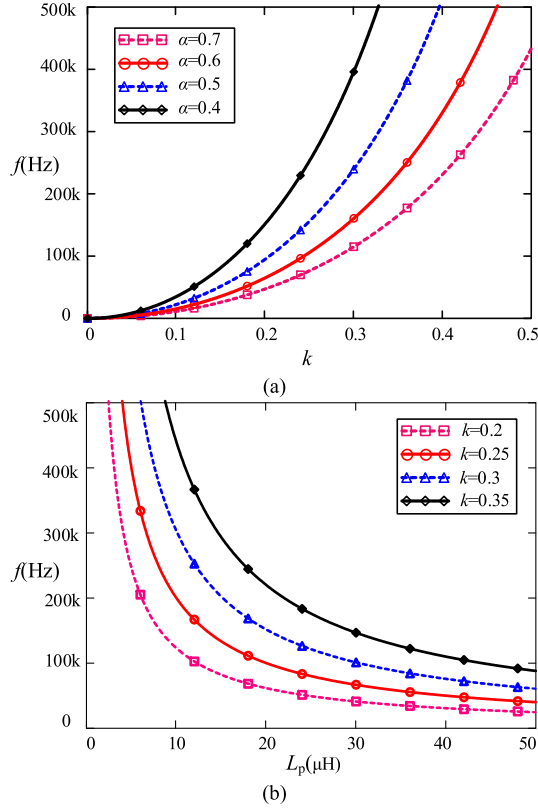


Fig. 14. Parameter characteristic curve. (a) Frequency f and the coupling coefficient k with different scale factors α . (b) Frequency f and the transmitting coil L_p with different coupling coefficients k .

and the value of L_p will decrease so as to achieve both CC and CV output. In order to present the relationship between f , α and L_p more intuitively, a 3-D map and a contour map about α are drawn, as shown in Fig. 15.

The calculation method of capacitance C_1 can be found in [29]. The value of ZVS_m can be given as needed to ensure reliable ZVS in both CC and CV modes. Fig. 16 shows ZVS_m of different C_1 values. It can be seen from Fig. 16 that with the growth of capacitor C_1 , both the blocking voltage of the switch and the output voltage gradually increase, reaching the critical state of ZVS at the capacitor C_1 of 190 nF. Therefore, with a given ZVS_m , the corresponding C_1 value can be selected. However, it will trigger the secondary resonance if ZVS_m is too large.

The voltage waveforms of compensation capacitors C_p and C_s in CC and CV modes are shown in Fig. 17. It can be seen from the waveforms that the voltage stress of the compensation capacitor C_s is basically the same in the CC and CV modes, which are both 117 V, but the voltage phase changes. The voltage stress of compensation capacitor C_p in CC and CV modes is 190 and 210 V, respectively, and the voltage phase does not change.

Taking the input voltage of the switched-capacitor-based Class-E inverter and the secondary rectifier as the base value, the stress voltages of C_p and C_s are normalized, which are 2.1 and 1.95, respectively.

The scale factor α satisfies the relationship shown in (30) and can be determined according to the desired input and

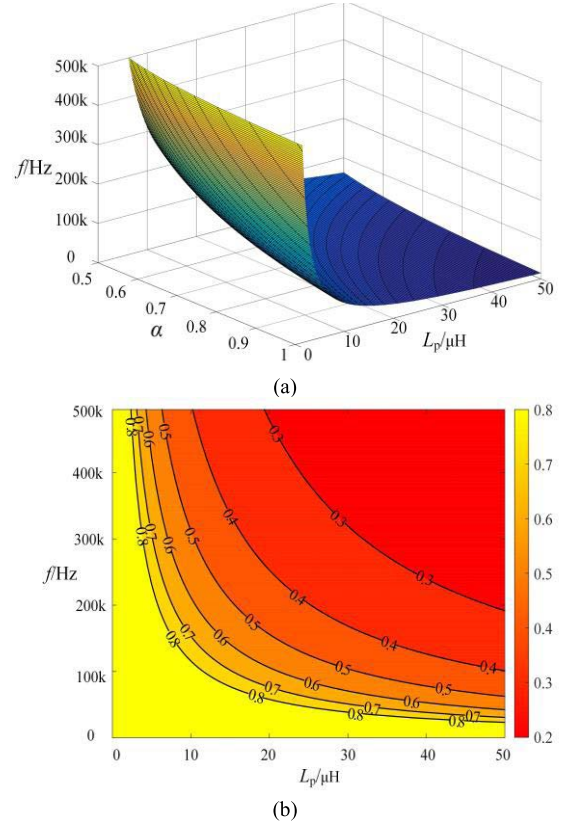


Fig. 15. (a) 3-D map with L_p , f , and α . (b) Contour map.

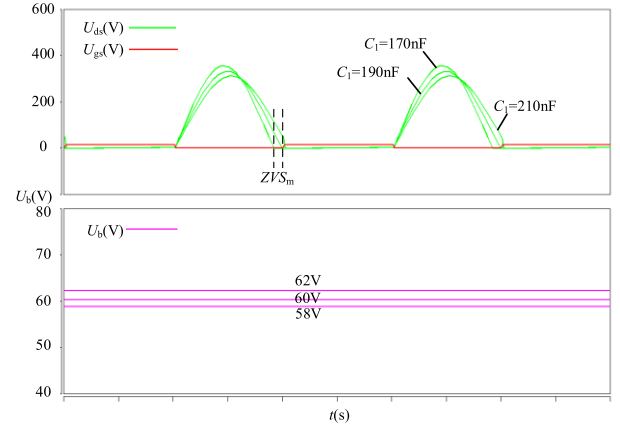


Fig. 16. ZVS_m of different C_1 values.

output conditions. The process of parameter design is shown in Fig. 18.

The first step is to determine the input voltage U_{dc} , battery charging voltage U_b , charging current I_b , frequency f , and desired ZVS_m . Second, the value of L_{p1} is determined according to the relationship between frequency and coupling coefficient. Then, the parameters C_p , C_{s1} , C_{s2} , L_2 , and A_1 can be calculated by (1)–(7), (18), and (19). Finally, the value of C_p is determined according to the value of A_1 .

Through the aforementioned process, the parameters of a 100-V-input Class-E circuit with 3-A output current in CC mode and 60-V output voltage in CV mode and its compensation network can be shown in Table II.

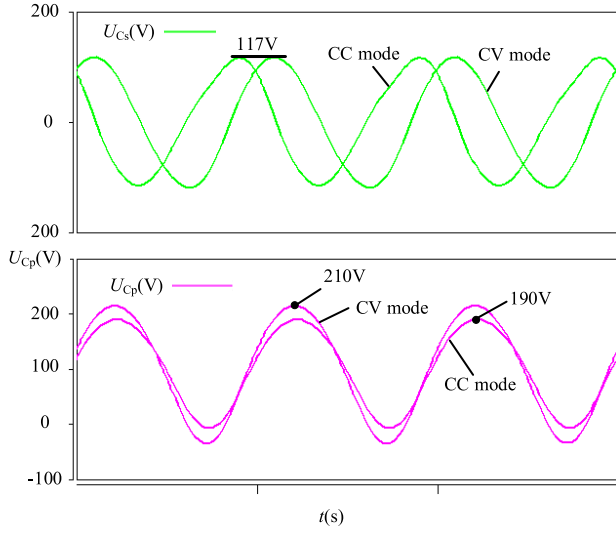


Fig. 17. Voltage waveforms of compensation capacitors.

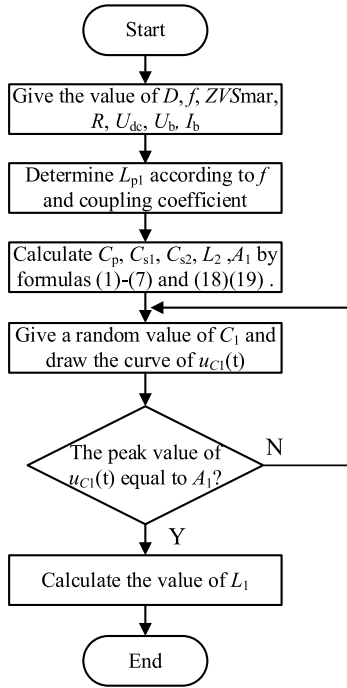


Fig. 18. Flowchart of parameter design.

In order to test the sensitivity of the system under these parameters, the sensitivity analysis of the circuit is carried out. The normalized output voltage in CV mode and the normalized output current in CC mode versus the normalized parameters of C_s , C_p , α , and ZVS_m are shown in Fig. 19. Obviously, the variation of ZVS_m is not sensitive to the output condition in both CC and CV modes. The value of C_s and C_p shares the same sensitivity to the change of CC output. Therefore, ZVS_m can be increased appropriately by reducing the value of C_1 so that ZVS can be reliably realized in both CC and CV modes.

IV. SIMULATION AND EXPERIMENT RESULTS

Using the parameters shown in Table II, the Saber-based simulation is performed, and the results are shown in Fig. 20.

TABLE II
CURRENT PARAMETERS OF CLASS-E CIRCUIT

Symbol	Definition	Value
f	frequency	100kHz
U_{dc}	Input dc voltage	100V
U_b	Output voltage	60V
I_b	Output current	3A
L_p	Transmitting coil inductance	20 μ H
L_s	Receiving coil inductance	30 μ H
M	Mutual inductance	6.12 μ H
α	Scale factor	0.616
C_p	Primary compensation capacitor	329.8nF
C_s	Compensation capacitor in CV mode	187.9nF
C_{s1}	Compensation capacitor in CC mode	94nF
ZVS_m	Zero voltage switching margin	8%
L_1	Primary side inductance	16 μ H
L_2	Secondary side inductance	26.9 μ H

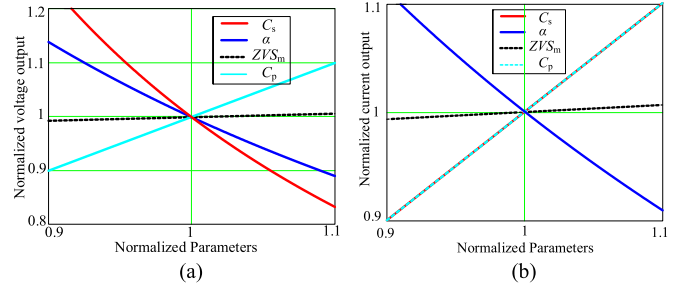


Fig. 19. Normalized output with varying normalized parameters. (a) CV mode and (b) CC mode of operation.

The waveforms from top to bottom are switch voltage u_{ds} and u_{gs} , output current i_b , transmitting coil current i_{Lp} , receiving coil current i_{Ls} , resonant network voltage waveform u_{C1} , and inductance voltage i_{L1} .

It can be seen from Fig. 20 that the switch achieves ZVS turn-on, the blocking voltage of the switch is 385 V, and the output meets the design requirements.

The waveforms of output voltage and current when the load is switched from full to half load are shown in Fig. 21(a) and (b), respectively. It can be seen from Fig. 21 that in the CV mode, when the load is switched from 10 to 20 Ω , the output voltage increased from 60.38 to 61.1 V, whose error is less than 1 V. In the CC mode, when the load is switched from 10 to 20 Ω , the output voltage decreased from 3.08 to 2.99 A, whose error is less than 0.1 A. Moreover, settling time in both modes is less than 10 ms.

To verify its feasibility, a 180-W wireless charging prototype with 3-A charging current and 60-V charging voltage based on the proposed circuit is built, as shown in Fig. 22. The platform consists of a transmitting circuit, a transmitting coil, ferrite shielding layer, a receiving coil, a receiving circuit, oscilloscope, and control circuit marked with 1–7 in turn. The measured circuit parameters are shown in Table III, and the other circuit parameters are the same as the simulation.

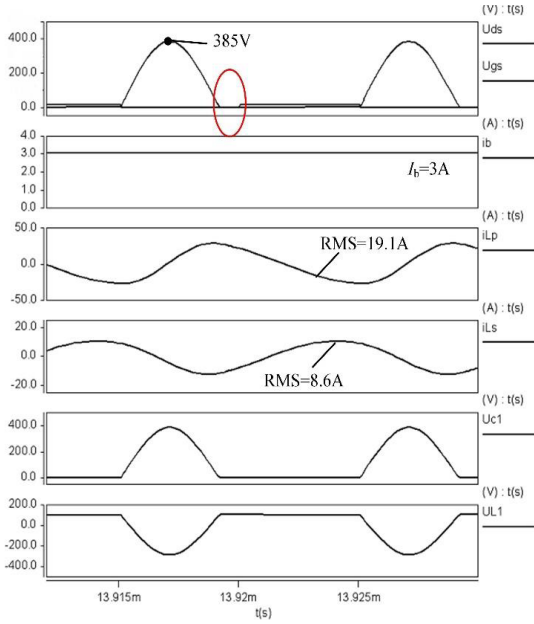


Fig. 20. Simulation waveforms of the Class-E circuit.

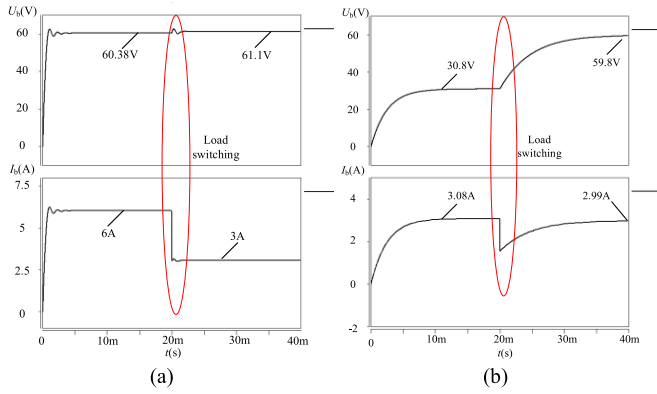


Fig. 21. Waveforms of output voltage and current. (a) Load switching diagram under CV mode. (b) Load switching diagram under CC mode.

Fig. 23(a) and (b) shows the ZVS waveforms in CV and CC modes, respectively. It can be seen from Fig. 23(a) that the measured amplitude of u_{ds} is 390 V, which is close to the simulation result of 385 V. The difference between the two values is caused by the inaccuracy of the actual compensation parameters and parasitic parameter. ZVS_m can be computed as $\Delta T \times f = 7.5\%$, which illustrates that the calculation method proposed in this article has high accuracy and reliability.

Fig. 24(a) and (b) shows the waveforms of output voltage u_o and receiving coil current i_{Ls} in the CV mode, when the resistive load is set as 20 and 40 Ω , respectively. The output voltage is maintained at 60 V even if the resistive load and i_{Ls} change in a wide range. Thus, the load-independent voltage output is achieved.

Fig. 25(a) and (b) shows the output current waveforms of the circuit in CC mode when the resistive load is set as 20 and 10 Ω , respectively. The output current is maintained stable at 3 A. Thus, the load-independent current output is achieved.

Fig. 26(a) and (b) shows the output waveforms when the power changes back and forth between full and half power in CC and CV modes, respectively.

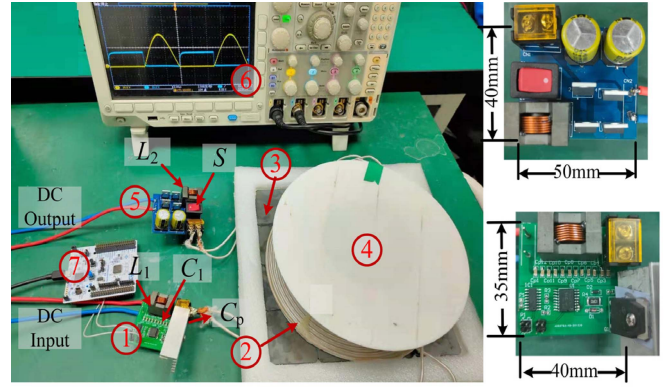


Fig. 22. Wireless charging platform based on the proposed circuit.

TABLE III
CALCULATED AND MEASURED PARAMETERS OF CLASS-E CIRCUIT

Parameter	Calculated	Measured
L_p	20 μ H	20.4 μ H
L_s	30 μ H	31.1 μ H
M	6.12 μ H	6.05 μ H
C_p	329.8nF	330nF
C_s	187.9nF	187.3nF
C_{s1}	94nF	93.6nF
L_1	16 μ H	16.2 μ H
L_2	26.9 μ H	24.5 μ H

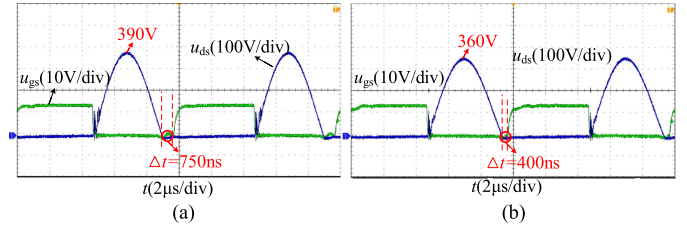
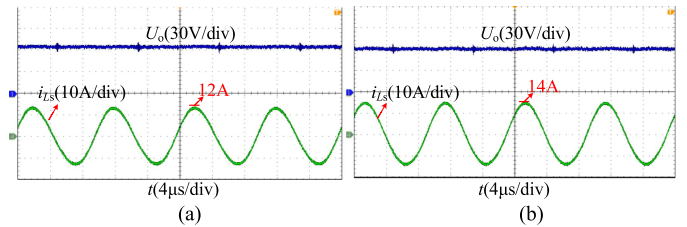


Fig. 23. Waveforms of ZVS in (a) CV and (b) CC modes.

Fig. 24. Output voltage waveforms of the circuit in CV mode. Resistive load is set as (a) 20 and (b) 10 Ω .

It can be seen from Fig. 26(a) that in the CC mode, the output current approximately remains constant and the output voltage is reduced to half when the load changes from 20 to 10 Ω . In the CV mode, as shown in Fig. 26(b), the output voltage approximately remains constant, and the output current is doubled.

Fig. 27 shows the dynamic characteristics of the system when the operation mode is switched from CC mode to CV mode at the load of 20 Ω . When the operation mode is

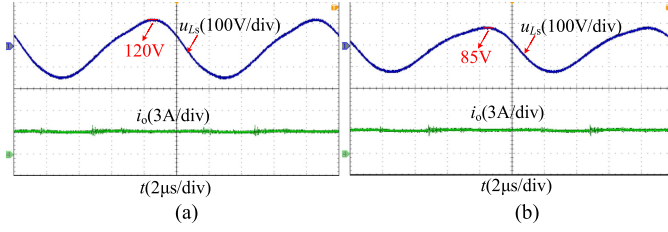


Fig. 25. Output current waveforms of the circuit in CC mode. Resistive load is set as (a) 20 and (b) 10 Ω .

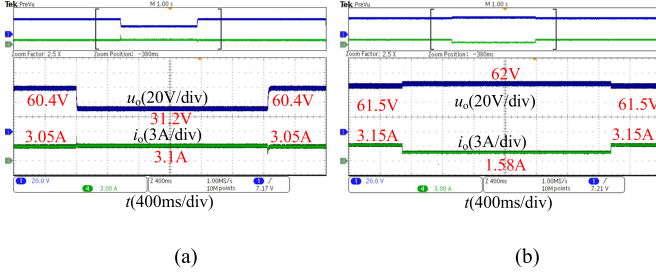


Fig. 26. Experimental waveforms of u_o and i_o . (a) CC mode with the load change from 20 to 10 to 20 Ω . (b) CV mode with the load change from 20 to 40 to 20 Ω .

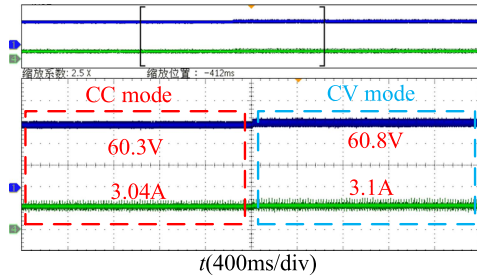


Fig. 27. Output voltage and current waveform from CC mode to CV mode.

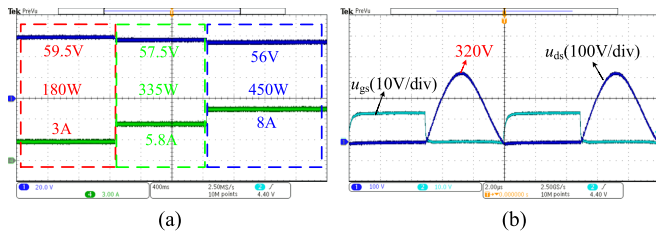


Fig. 28. (a) Output voltage and current when the load changes from 20 to 10 to 7 Ω . (b) ZVS waveform when the load is 7 Ω .

switched from CC to CV, both the output voltage and current vary slightly.

In addition, in order to further expand the application range of the circuit, the power of the circuit can be increased to 450 W on the premise of ZVS.

Fig. 28(a) shows the output voltage and current when the load changes from 20 to 10 to 7 in CV mode. It can be seen that the output power of the circuit is about 450 W when the load is 7 Ω , and the ZVS operation can be achieved with 450-W output power, as shown in Fig. 28(b).

The actual Fourier decomposition of the resonant circuit voltage U_{in} is shown in Fig. 29.

It can be seen from Fig. 30 that the measured higher harmonics coincide well with the theoretical values.

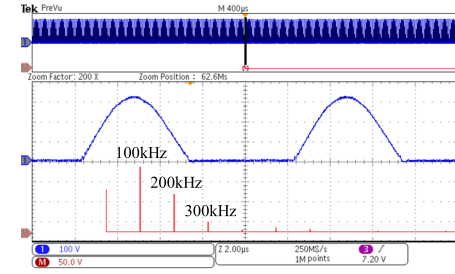


Fig. 29. FFT waveform of the voltage U_{in} .

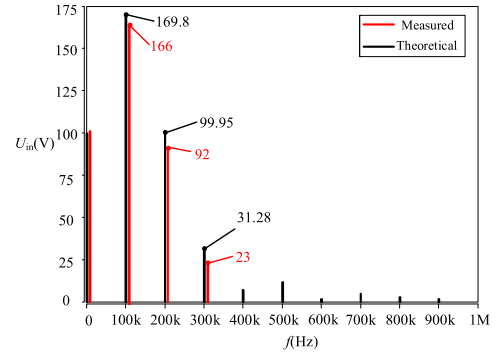


Fig. 30. Fourier decomposition diagram of the voltage U_{in} .

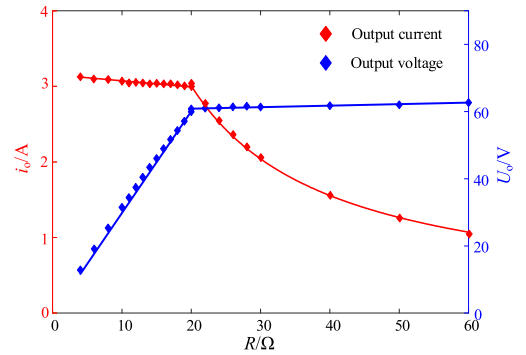


Fig. 31. Output voltage and current under difference battery equivalent resistance.

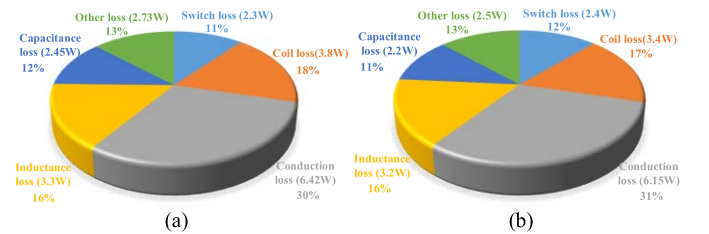


Fig. 32. Loss of the circuit at the load of 20 Ω . (a) Loss of CC mode. (b) Loss of CV mode.

During the charging process, the battery equivalent resistance rises up, and the output voltage and current under difference battery equivalent resistance are shown in Fig. 31.

The detailed analysis of circuit power loss in both CC and CV modes is shown in Fig. 32. It can be seen that the loss of the circuit before and after switching has little change, and the conduction loss of the switch is the largest, which is higher than the magnetic loss of the inductor.

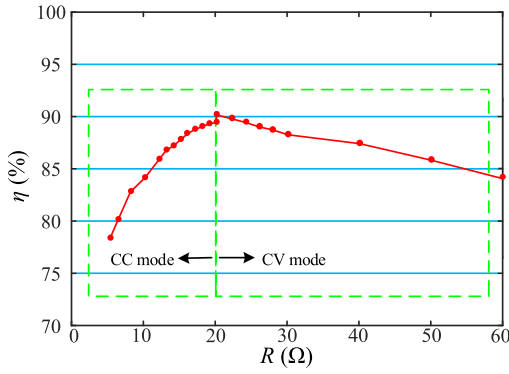


Fig. 33. Diagram of efficiency versus the load.

TABLE IV
PERFORMANCE COMPARISON OF CLASS-E CIRCUIT

Circuit	Frequency	Output Power	Efficiency	CC/CV
Proposed	100kHz	450W	90.3%	Yes
[31]	200kHz	24W	90%	No
[32]	200kHz	16W	85%	No
[14]	6.78MHz	30W	80%	Yes
[33]	6.78MHz	70W	90%	No
[34]	1MHz	3W	79.6%	No
[35]	1MHz	27.8W	74.7%	No
[36]	13.56MHz	25.6W	73.4%	No

Finally, the curve of efficiency from dc power source to resistive load processed is shown in Fig. 33. The results show that the maximum efficiency in CC mode is 89.5%, and the maximum efficiency in CV mode is shown as 90.3%. The efficiency improves because the output power increases when the system is switched from CC mode to CV mode.

The performance of the work in this article is compared with other existing Class-E circuit-based WPT, as shown in Table IV.

It can be seen that the proposed switched-capacitor-based Class-E inverter can achieve higher output power and efficiency compared with the traditional Class-E circuit listed in the table.

V. CONCLUSION

A new switched-capacitor-based Class-E topology circuit with S/CL compensation network has been proposed in this article to solve the problem of operating power range limitation with the traditional Class-E circuit, which also requires more components and complex control to realize CC and CV output modes. With the proposed topology circuit, soft switching can be realized and a simpler switch between CC and CV output modes can be achieved by only switching the capacitor branch once. The characteristics of the circuit have been analyzed in detail, and the compensation method of artificially decomposing the transmitting coil has been adopted to improve the design freedom. The design method studied in this article can be extended to other WPT systems with different topology circuits. Besides, this article has introduced the variable ZVS_m and analyzed the sensitivity of parameters and the influence of

higher harmonics on the circuit, which makes the theoretical analysis and parameter calculation more accurate. Finally, a 180-W prototype has been built to validate the effectiveness of the proposed Class-E topology circuit with CC and CV output modes at different loads and to show an excellent agreement with the theoretical analysis.

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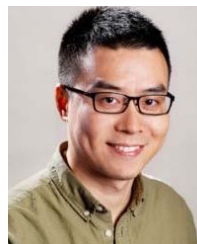
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